Poly-Kritter

Virtual analog polysynth · CLAP plugin and web instrument



Poly-Kritter is a polyphonic synthesizer built from Kritter's voice: two oscillators, a ring modulator and noise, a high-pass and a low-pass filter in series, a two-slider envelope and a VCA, with a shared vibrato and a repeat clock. Up to sixteen voices play at once, one to a note or stacked up to sixteen on one, with an eight-route modulation matrix, chorus, delay, reverb and a limiter.

This manual describes every control. Its pictures of the faceplate are drawn by the plugin's own code, and its graphs are measured by running the plugin's engine. The example patches in the Patches chapter are factory presets: load them by name and play.

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Overview

What it is

Kritter is a monophonic synthesizer modelled on one from the 1970s. Poly-Kritter gives every note its own copy of Kritter's voice, all under one panel, and adds what a keyboard instrument needs: voice allocation and stacking, a modulation matrix, effects and presets.

Kritter's panel works in its own terms rather than an ADSR's. The envelope has two sliders and two switches; vibrato, the Bender and portamento move the pitch in hertz, not in semitones; and the filter is two filters, a high-pass and a low-pass, that together pass a band. The rest of this chapter is the map; the sections after it take the faceplate a plate at a time.

This is the manual for the CLAP plugin and the web build, which are the same engine and the same faceplate. Poly-Kritter for Voltage Modular, with jacks, has a manual of its own.

Signal path

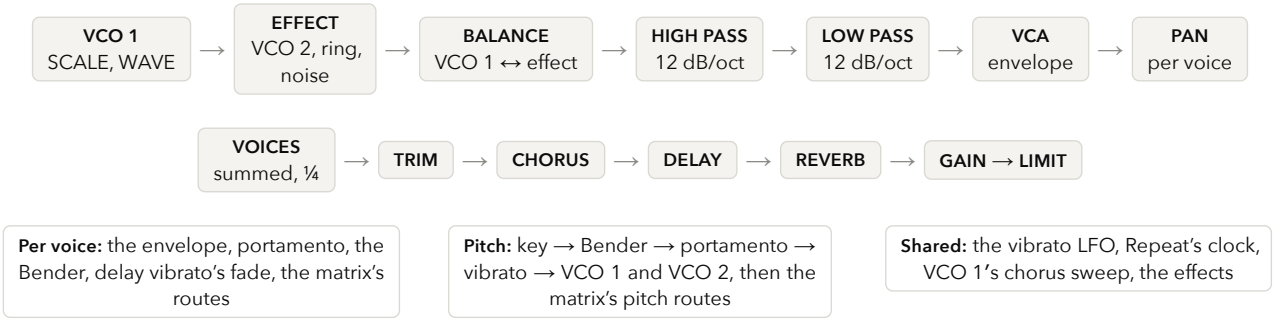


Figure 1. One voice, then the shared chain. The oscillators, the effect section, the filters and the VCA run at twice the host's sample rate; the effects at the host's rate.

Each voice has its own envelope, so its notes start and end on their own; but there is one vibrato LFO and one Repeat clock for all of them, so every voice's vibrato moves in step and every repeating note pulses together.

The faceplate



Figure 2. The faceplate. Each number is the section of this manual that describes it.

1 **VCO 1**	the oscillator: octave and waveform	2 **EFFECT**	VCO 2, the ring modulator, noise
3 **FILTER**	high-pass and low-pass	4 **ENVELOPE**	two sliders, two switches
5 **VIBRATO**	the LFO and its three switches	6 **REPEAT**	re-strikes held notes
7 **GLIDE**	portamento and the Bender	8 **VOICES**	Poly, Solo; stacking
9 **MOD MATRIX**	opens the routing	10 **FX**	trim, chorus, delay, reverb
11 **OUTPUT, scope**	gain, limiter, level	12 **Preset bar, RAND**	presets, and a new random sound

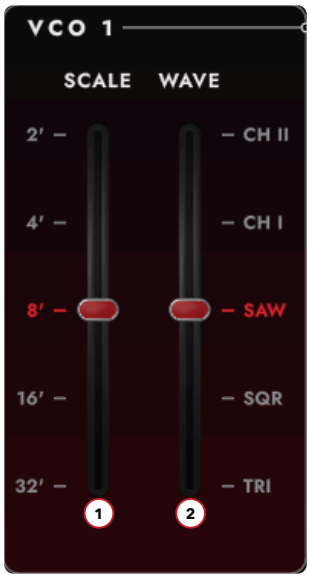
Faders read upwards, and the switches have ON at the top. The red mark beside a control is its setting at Init. Knobs turn by dragging up and down, or by scrolling. Hover over a control to see its value; the faders show theirs in hertz, seconds or semitones. Shift-drag for fine adjustment; double-click, or Cmd-click (Ctrl-click on Windows), returns a control to its Init setting.

Init

A new instance starts from the Init patch, Kritter's panel at rest: one sawtooth at 8', both filters open, the envelope's attack fast and its level held for as long as the key is, every switch off but EFFECT and STOP. It is an organ: notes start and stop with the keys. Velocity does nothing until the matrix routes it.

On the strip below, each note is one voice (**WIDTH** 1), **POLY**, and the matrix's slot 1 routes **Spread** to **Pan**, which does nothing until a stack is spread. Every effect and the limiter are off, and **GAIN** is at unity. **RESET** returns to Init.

1 · VCO 1



1. **SCALE** octave: 32' to 2'
2. **WAVE** TRI, SQR, SAW, CH I, CH II

SCALE selects VCO 1's octave in five steps, 32', 16', 8', 4' and 2', with 2' at the top. At 8', where it starts, each key plays its own pitch: A4 is 440 Hz. Each step is an octave.

SCALE also moves both filters' cutoffs by an octave a step, so a filter setting keeps the same relation to the notes when the octave changes. The filters do not follow the key played, only this switch; for that, route **Key** to **Cutoff** in the matrix.

WAVE selects the waveform. It starts at SAW.

Position	Waveform
TRI	Triangle. Odd harmonics only, falling at 12 dB per octave ($1/n^2$).
SQR	Square. Odd harmonics only, falling at 6 dB per octave ($1/n$).
SAW	Sawtooth. Every harmonic, falling at 6 dB per octave ($1/n$).
CH I	Pulse, its width swept by a 0.8 Hz triangle between 15% and 85%.
CH II	Pulse, its width swept by a 5 Hz triangle between 40% and 60%.

The chorus sweep is one oscillator shared by every voice, so a chord's pulse widths move together. It runs whether or not a note is playing, at CH I's rate unless WAVE is at CH II. The matrix can read it as the source **Chorus LFO** whatever the waveform. There is no sine wave.

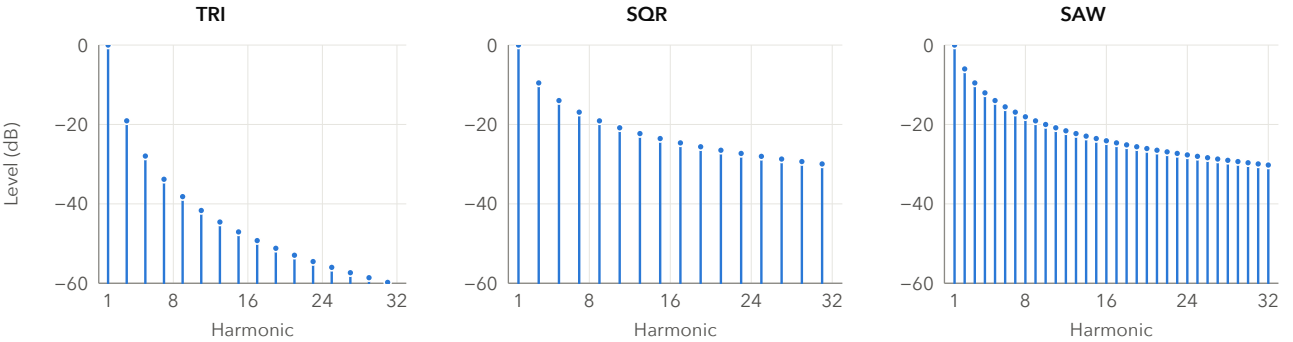
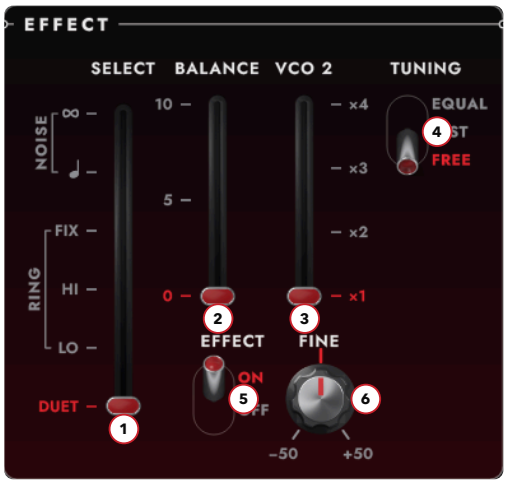


Figure 3. Harmonic levels against the fundamental, measured at the output with the filter open, C3 (130.8 Hz). The even harmonics of TRI and SQR are below −120 dB.

WAVE	RMS (dBFS)	Peak (dBFS)
TRI	−16.8	−12.3
SQR	−12.1	−7.3
SAW	−16.8	−7.9
CH I	−13	−5
CH II	−12.1	−6.4

Table 1. One voice's level at the output: C3, the filter open, **GAIN** at unity, every effect off. The peaks are higher than the waveforms' shapes suggest because the band-limiting and the high-pass at its lowest shift the harmonics' phases; the RMS level is the better guide to loudness.

2 · EFFECT



1. **SELECT** — the section's source
2. **BALANCE** — VCO 1 against the effect
3. **VCO 2** — VCO 2's ratio to VCO 1, ×1 to ×4
4. **TUNING** — FREE, JUST, EQUAL
5. **EFFECT** — the section on or off
6. **FINE** — VCO 2's fine tuning, ±50 cents

The effect section holds a second oscillator, a ring modulator and a noise source. **SELECT** chooses what it produces, and **BALANCE** mixes that with VCO 1. With BALANCE at 0, where it starts, the section is not heard.

SELECT has six positions, DUET at the foot, where it starts:

Position	Output
DUET	VCO 2, a sawtooth, following the key at the ratio set by VCO 2.
RING LO	VCO 1 ring-modulated by VCO 2, VCO 2 following the key as at DUET.
RING HI	As RING LO, with VCO 2 two octaves higher.
RING FIX	VCO 1 ring-modulated by VCO 2 at a fixed frequency, not following the key.
NOISE ♩	Noise ring-modulated by VCO 2, VCO 2 following the key two octaves up: noise with a pitch.
NOISE ∞	White noise.

The selector switches without a crossfade, so moving it during a note clicks.

The ring modulator

The ring modulator switches the polarity of its input at VCO 2's frequency: VCO 2 is clipped into a square with rounded edges, and the input is multiplied by it. The output holds the sums and differences of the input's frequencies and the square's odd harmonics, and neither the input's own frequencies nor VCO 2's. Only VCO 2's zero crossings matter, so its waveform does not.

With VCO 2 following the key (RING LO, RING HI), the sidebands keep the same relation to every note, and the result plays in tune. At RING FIX VCO 2 stays put while VCO 1 follows the key, so each note makes a different, inharmonic set of sidebands: bells, gongs and clangs. The fixed carrier is set by VCO 2 and FINE, from 110 Hz at ×1 to 440 Hz at ×4.

BALANCE crossfades linearly from VCO 1 alone at 0 to the effect section alone at 10; at 5 the two are equal. In DUET it sets VCO 2's level against VCO 1; at the ring and noise positions it sets how much plain VCO 1 remains. The matrix's **Balance** destination adds to it.

VCO 2 sets VCO 2's frequency as a ratio of VCO 1's at 8', from ×1 (unison) to ×4 (two octaves up), linear in frequency: ×2 is a third of the way up. VCO 2 counts from 8' whatever SCALE is set to, so at 16' the two are an octave apart with the fader at ×1. It starts at ×1.

TUNING sets how the fader is read. **FREE** is continuous. **JUST** snaps it to the nearest whole-number ratio: unison, the thirds, fourth, fifth, sixths and sevenths (5/4, 3/2, 5/3 …), the octave, the same an octave higher, and ×4. **EQUAL** snaps it to the nearest equal-tempered semitone. The fader's tooltip names the interval, "×1.500, 5th (3/2)". It starts at FREE.

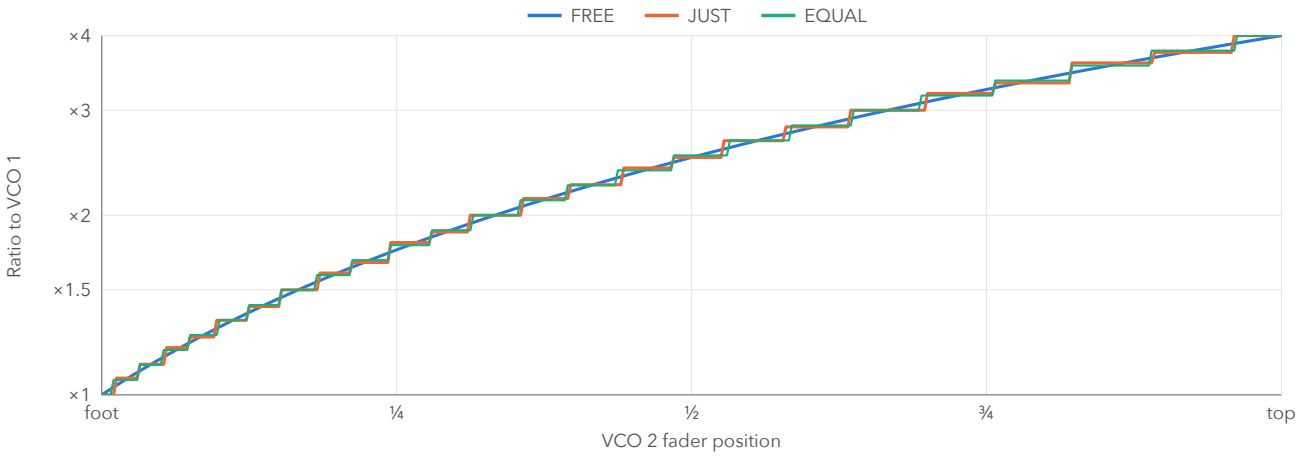


Figure 4. VCO 2's ratio against the fader under each TUNING, FINE centred. The scale is logarithmic, so equal heights are equal intervals.

FINE (−50 to +50 cents) detunes VCO 2 from wherever the fader lands, under every TUNING. It starts at its centre. At ×1 with FINE centred, VCO 1 and VCO 2 are identical, so in DUET with VCO 1 on SAW, BALANCE changes nothing; detune FINE by a few cents to hear the two beat.

EFFECT switches the section on and off. Off, the voice plays VCO 1 alone whatever BALANCE is set to. It starts on.

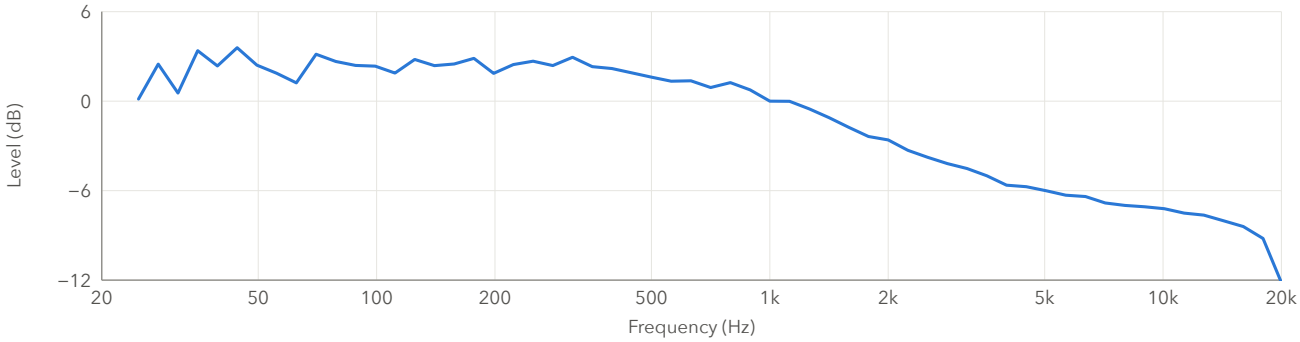


Figure 5. NOISE ∞ at the output, in sixth-octave bands, against the band at 1 kHz. The noise source's own filtering tilts it down by about 10 dB from the bass to 16 kHz.

3 · FILTER



1. **HIGH PASS** high-pass cutoff, 0 to 10
2. **LOW PASS** low-pass cutoff, 0 to 10
3. **STOP** keeps LOW PASS at or above HIGH PASS
4. **RESO** resonance of both filters
5. **ENV** the envelope's sweep of both cutoffs

The filter is a 12 dB/oct high-pass followed by a 12 dB/oct low-pass, each a two-pole diode-ring filter with its own fader. Together they pass a band: moving both faders together moves the band, and moving them apart widens it.

HIGH PASS and **LOW PASS** (0 to 10) set the two cutoffs, an octave a mark:

Mark	0	1	2	3	4	5	6	7	8	9	10
Cutoff (Hz)	20	40	80	160	320	640	1280	2560	5120	10240	20480

The figures are for SCALE at 8'; each step of SCALE moves both cutoffs an octave. The faders start at HIGH PASS 0 and LOW PASS 10, which leaves the audio band open.

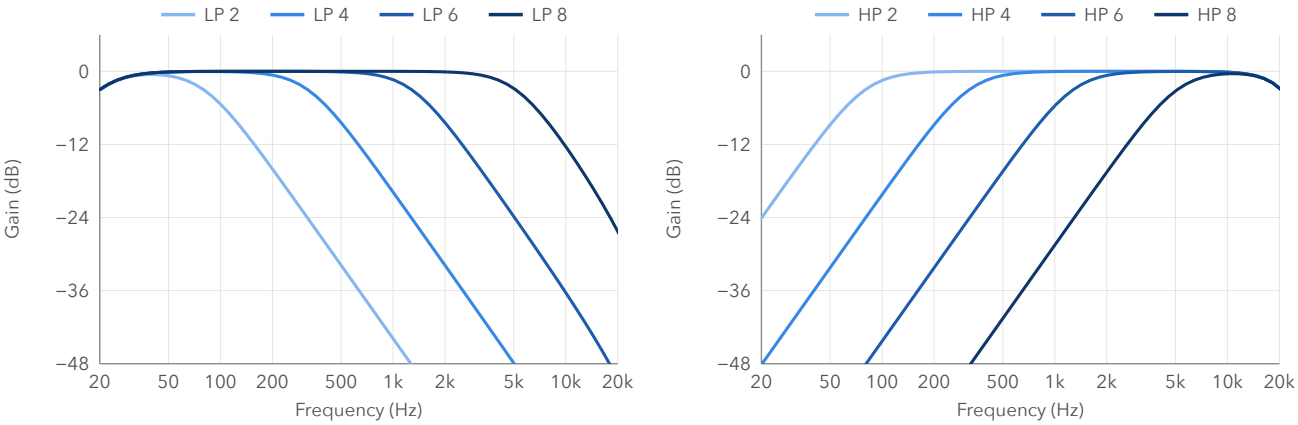


Figure 6. Left: LOW PASS at four settings, HIGH PASS at 0. Right: HIGH PASS at four settings, LOW PASS at 10. RESO off. At each cutoff the gain is −3 dB, and an octave beyond it −12 dB.

The diode rings compress large signals smoothly rather than clipping them; with RESO off the effect is small.

RESO (OFF to ON) sets both filters' resonance together. At OFF each has a Q of 0.71, the flattest response without a peak. At ON the low-pass rings for about five cycles and the high-pass for about four. The passband level does not change; the response rises at each cutoff. It starts at OFF.

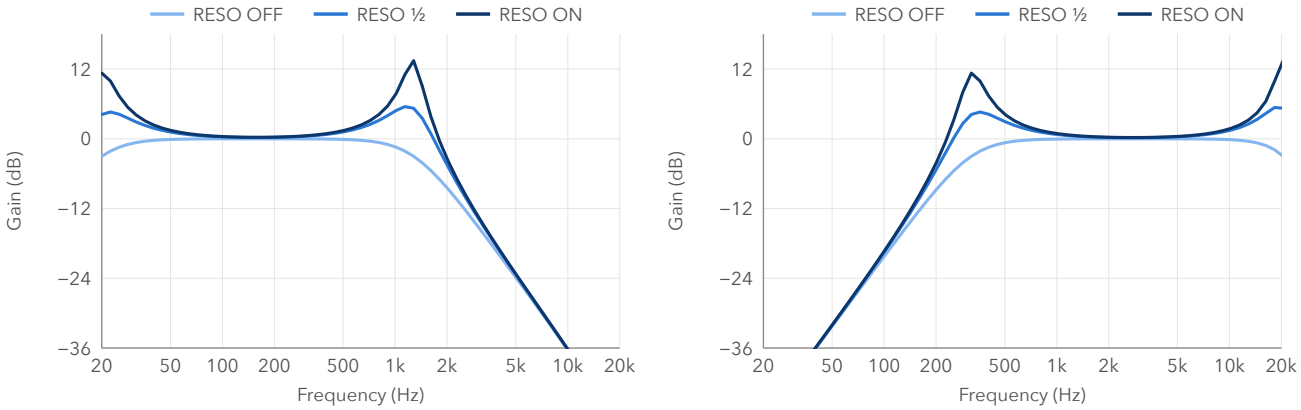


Figure 7. RESO at three settings. Left: LOW PASS at 6, HIGH PASS at 0. Right: HIGH PASS at 4, LOW PASS at 10. RESO acts on both filters, so each graph also shows the other filter's peak at the edge of the band. Small-signal measurements.

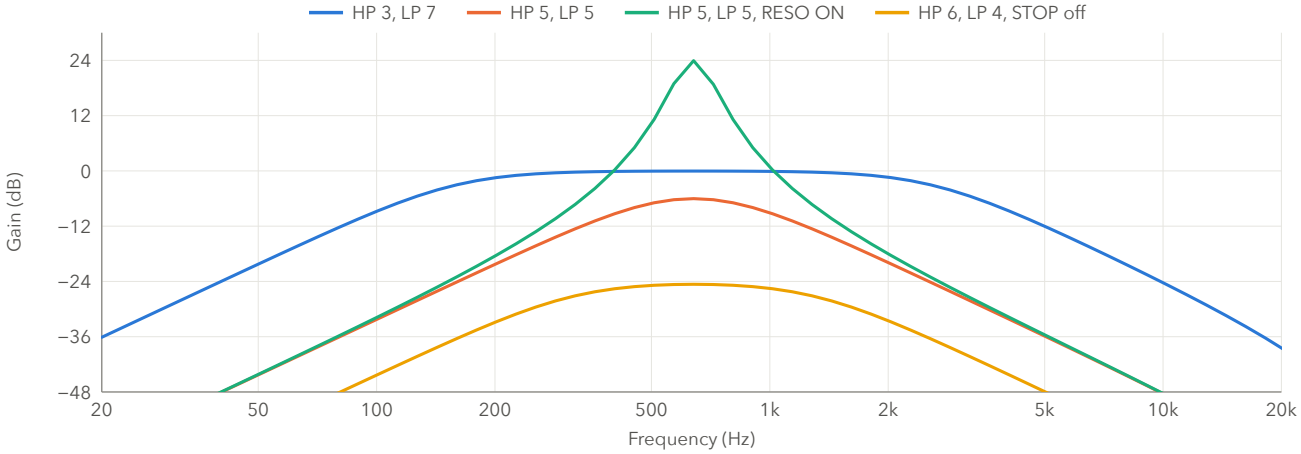


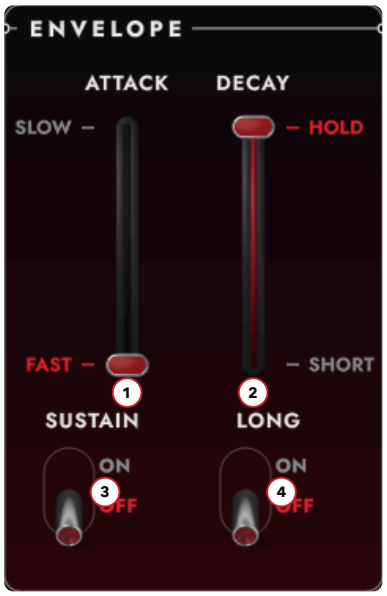
Figure 8. The two filters together. With both cutoffs at one setting the band narrows to a peak −6 dB; with RESO on, the two resonances add to +23.9 dB. With STOP off and LOW PASS below HIGH PASS, the slopes overlap and the level falls by about 25 dB.

STOP keeps the low-pass at or above the high-pass. On, if LOW PASS is set below HIGH PASS, the low-pass is held at HIGH PASS. It starts on. Off, the two are independent, and LOW PASS below HIGH PASS closes the band (the figure above). Matrix routes to **Low Pass** and **High Pass** count as part of the faders' settings, so STOP applies to them; ENV, CUTOFF vibrato, SCALE and the **Cutoff** route move both cutoffs alike and cannot break the rule.

ENV (OFF to ON) sets how far each note's envelope sweeps both cutoffs upwards: up to four octaves at the envelope's peak at ON, proportionally less between. The ATTACK and DECAY that shape the level therefore shape the tone. It starts at OFF.

The cutoffs move smoothly: fader and matrix changes take about 5 ms to arrive.

4 · ENVELOPE



1. **ATTACK** rise time, FAST to SLOW
2. **DECAY** what a held note does, SHORT to HOLD
3. **SUSTAIN** a long release after the key
4. **LONG** attack, decay and SUSTAIN eleven times longer

The envelope has two faders, not four, and it is not an ADSR: there is no sustain level. While the key is held, the envelope rises to its peak and then either holds there or decays towards silence. When the key is let go it falls: quickly, or slowly with **SUSTAIN** on. The envelope drives the VCA, and the filter through **ENV**.

Each segment is the charge or discharge of a capacitor, an exponential curve towards a target. A new note on a voice starts its attack from the envelope's current level, so a note played into another's release does not drop to zero first.

ATTACK (FAST to SLOW) sets the rise from zero to the peak: 2 ms at FAST to 1 s at SLOW, exponentially along the fader. The rise aims at 1.5 times the peak and stops at the peak, so it is fast at first and slows into the top. It starts at FAST.

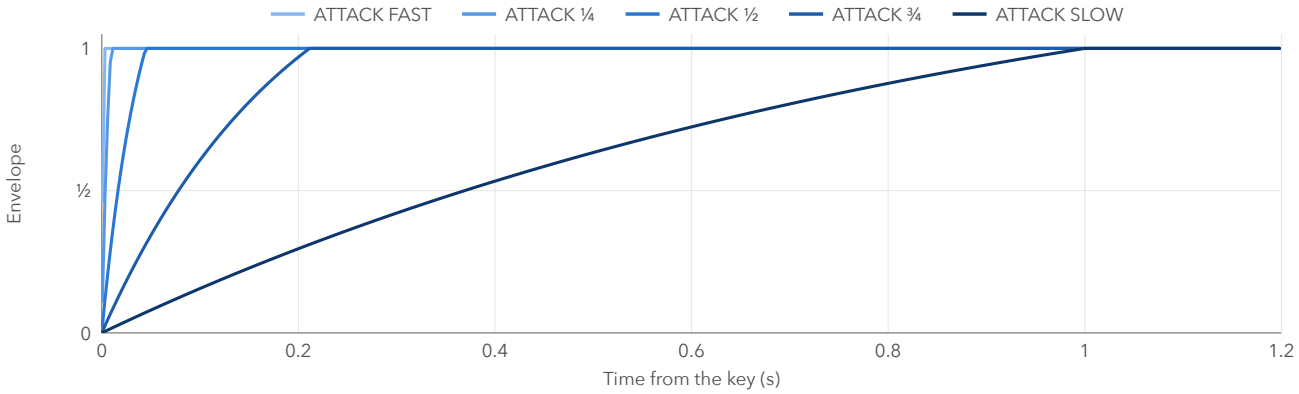


Figure 9. The attack at five ATTACK positions, DECAY at HOLD, measured on one voice.

DECAY (SHORT to HOLD) sets what a held note does after the attack. At HOLD, where it starts, the envelope stays at its peak for as long as the key is held. Below HOLD it decays towards zero, with a time constant of 0.15 s at SHORT that lengthens without limit towards HOLD: at SHORT a held note falls 40 dB in 0.7 s.

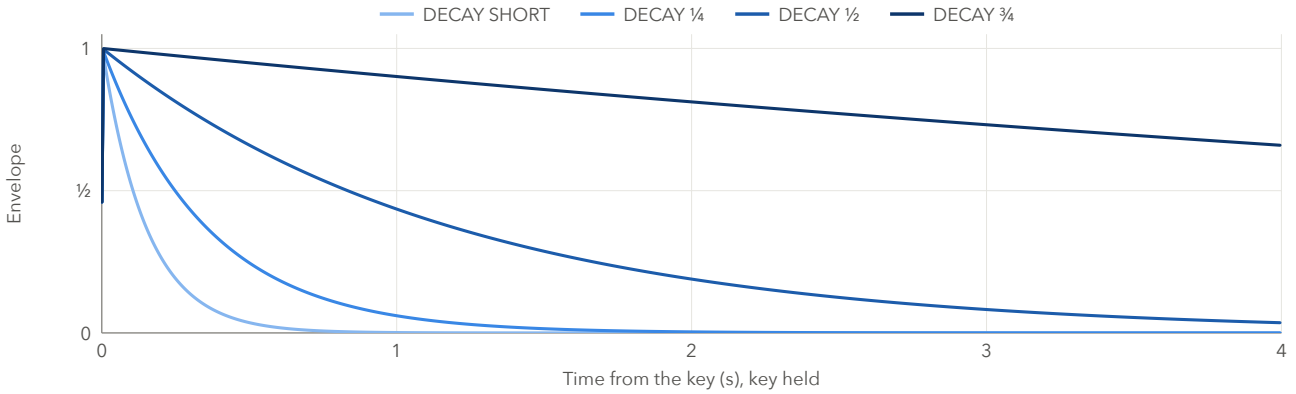


Figure 10. The held decay at four DECAY positions, ATTACK at FAST. Time constants 0.15 s, 0.36 s, 1.2 s and 9.6 s.

SUSTAIN sets the release, and only the release. Off, where it starts, the envelope falls with a 10 ms time constant when the key is let go, and the VCA shuts 47 ms after the key: fast, but without a click. On, the time constant is 0.3 s, and the note is heard for about 1.4 s after the key.

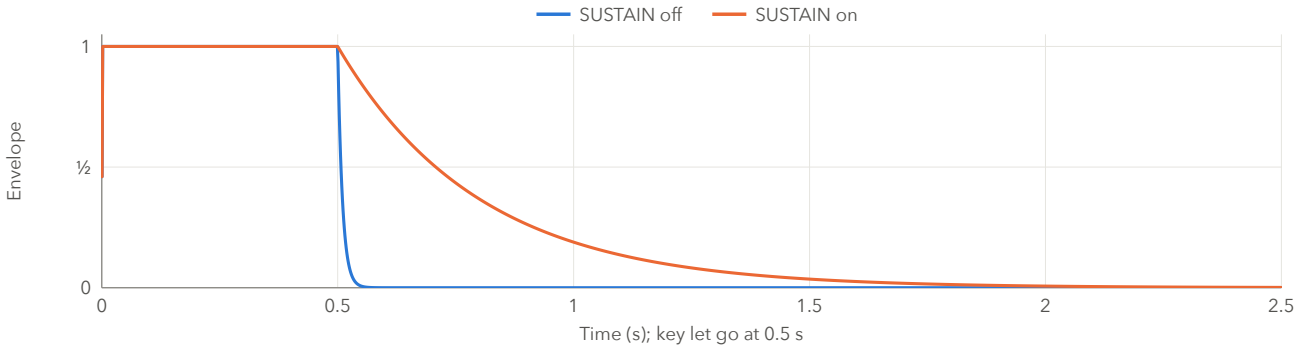


Figure 11. The release, ATTACK at FAST and DECAY at HOLD.

LONG multiplies the attack, the held decay and the SUSTAIN release by eleven. At SLOW the attack takes 11 s; with SUSTAIN on the release is heard for about 15 s. It does not lengthen the short release, so with SUSTAIN off a note still stops with the key. It starts off.

The VCA follows the envelope linearly, and shuts completely when the envelope falls below 1% of its peak (−40 dB), so a released note ends in silence rather than a long faint tail.

Sound	ATTACK	DECAY	SUSTAIN
Organ	FAST	HOLD	off
Plucked string, harpsichord	FAST	¼ up from SHORT	off
Struck bar	FAST	SHORT	on
Flute, bowed string	just under ½	HOLD	off
Bass with a ring after the key	FAST	¼ up from SHORT	on
Chime	FAST	most of the way to HOLD	on, with LONG

The matrix can move the envelope's times per note: **Attack**, **Decay** and **Env Length** (both) add to the faders' positions, so **Velocity** into **Attack** at a negative depth makes hard notes start faster (Section 9). By default velocity does not touch the level; route **Velocity** to **Amp** for that, scaled by **Env**.

5 · VIBRATO



1. **SPEED** the LFO's rate, 2 to 15 Hz
2. **DEPTH** depth of vibrato and CUTOFF
3. **VIBRATO** vibrato from the start of each note
4. **DELAY** vibrato that fades in
5. **CUTOFF** the LFO on both filter cutoffs

One triangle LFO, shared by every voice, and three switches that send it to the pitch or the filter. The matrix reads it as the source **LFO**.

SPEED (SLOW to FAST) sets the rate, 2 Hz to 15 Hz, exponentially along the fader; 5.5 Hz at the centre, where it starts.

DEPTH (0 to 10) sets the depth of the vibrato and of CUTOFF. It has an audio taper: its centre, where it starts, is a tenth of full depth, so the lower half gives fine control of ordinary vibrato and the upper half large depths. The matrix's **Vibrato Depth** adds to it per voice, so a patch with DEPTH at 0 and VIBRATO on has vibrato only when the mod wheel or aftertouch asks.

VIBRATO applies vibrato to the pitch from the start of each note. **DELAY** fades it in over 0.6 s from each new note instead; the fade advances only while the key is held, and a legato note does not restart it. With both on, VIBRATO wins: full depth from the start. Both start off.

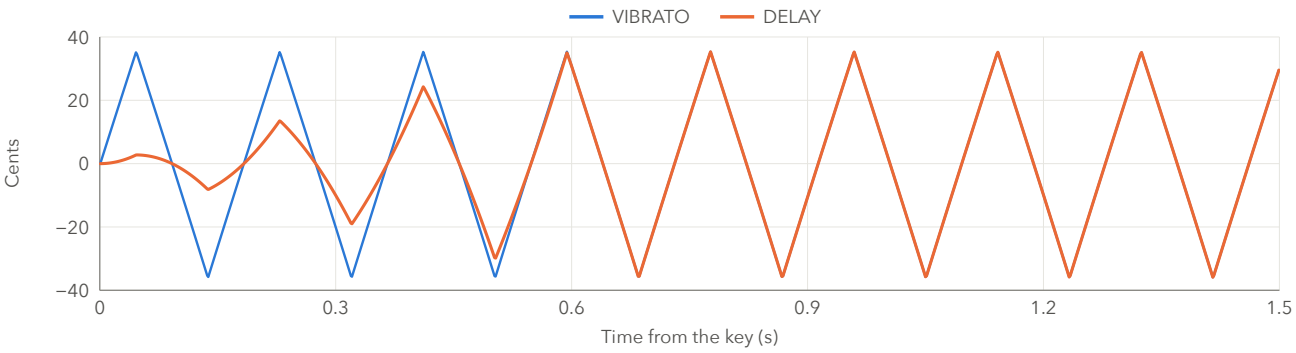


Figure 12. C3's pitch under VIBRATO and under DELAY, DEPTH and SPEED at their centres.

The vibrato is a fixed deviation in hertz, not in pitch, so it is wider in cents on low notes than on high ones. At the centre of DEPTH it is ±2.7 Hz: ±71 cents at C2, ±36 at C3, ±9 at C5. Section 14 explains why.

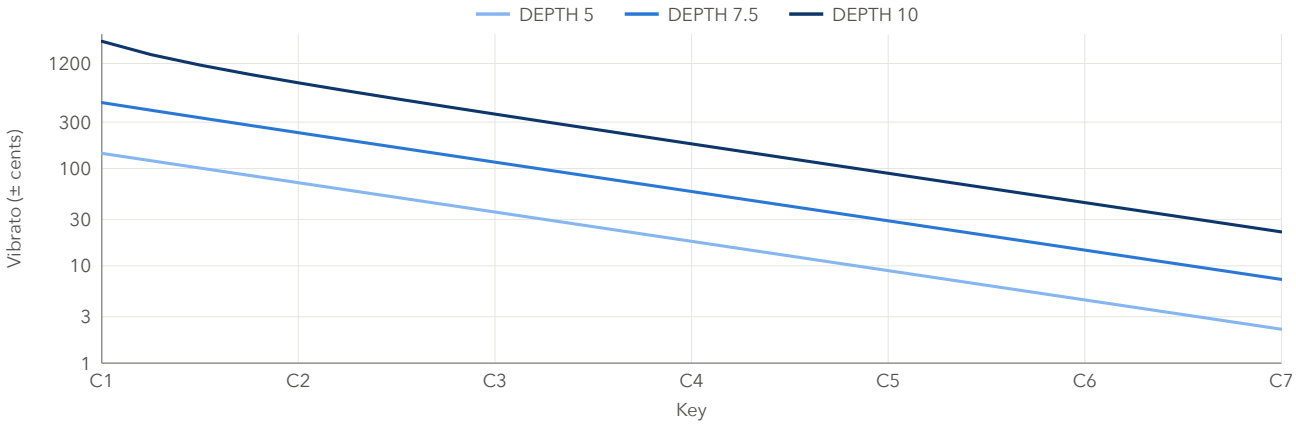
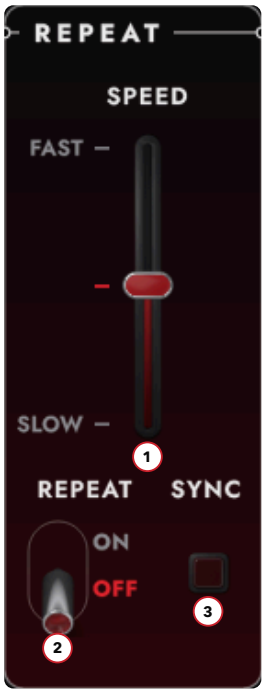


Figure 13. Vibrato's swing across the keyboard at three DEPTH settings, measured on a voice's pitch at 8′. It halves with each octave up; at full DEPTH on the lowest notes it passes an octave.

CUTOFF applies the LFO to both filter cutoffs, so the band moves up and down at SPEED: ±0.4 octave at the centre of DEPTH, ±4 octaves at full. It works without VIBRATO or DELAY and is never delayed. It starts off.

6 · REPEAT



1. **SPEED** the clock's rate, 2 to 25 Hz
2. **REPEAT** re-strikes held notes
3. **SYNC** SPEED as a division of the tempo

Repeat is a clock that re-strikes held notes at a steady rate. There is one clock for every voice.

SPEED (SLOW to FAST) sets the clock's rate, 2 to 25 Hz, exponentially along the fader; 7.1 Hz at the centre, where it starts.

REPEAT turns re-striking on. While a key is held, the clock opens the envelope's gate for the first half of each period and closes it for the second, so each period is a new attack and each gap a release. It starts off.

The clock is restarted by the first note of a phrase: a key that goes down while every key is up. Its first gap comes half a period after that key, and its first re-strike a period after it. While any key is held the phrase continues, and a note added to it leaves the clock running: it sounds from its key until the clock's next period begins, then follows the clock, joining the pulse of the notes already held. The phrase ends when every key is up.

Because Repeat works through the envelope, the envelope decides what it sounds like. With ATTACK at FAST and SUSTAIN off each strike is sharp and each gap silent. With SUSTAIN on the gaps fill in, and the strikes become a pulse on a held note. With a slow attack and a fast SPEED the envelope does not reach its peak before the next strike.

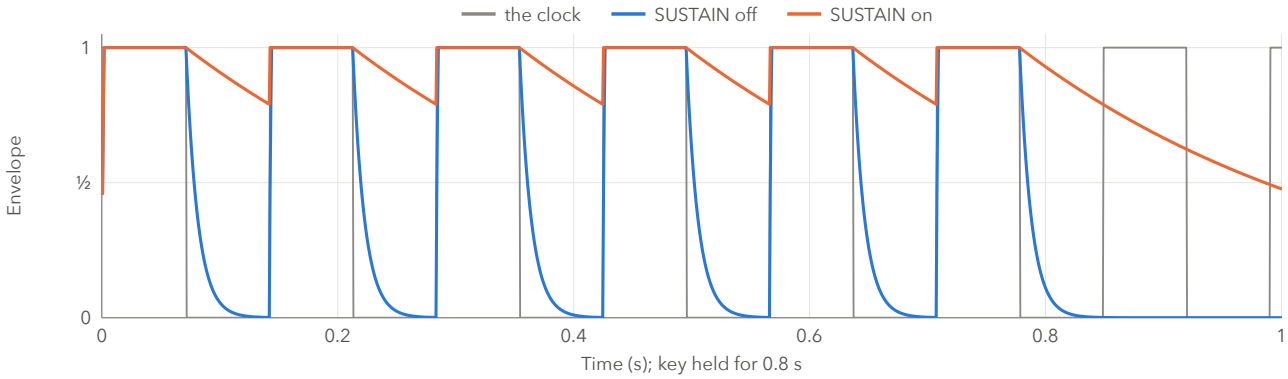


Figure 14. Repeat at SPEED's centre (7.1 Hz), ATTACK at FAST, DECAY at HOLD: the clock, and a voice's envelope.

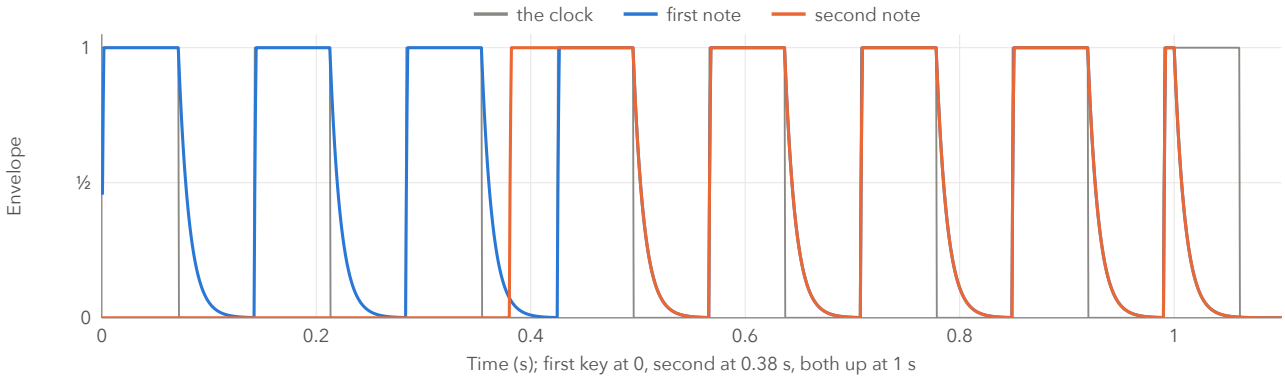


Figure 15. A phrase of two notes, SUSTAIN off. The second key falls in one of the clock's gaps: it sounds at once, holds until the next period begins, then repeats with the first.

SYNC locks the clock to the host's tempo. With SYNC on, SPEED picks a note value in twenty equal steps, from four bars at the foot to a thirty-second-note triplet at the top, and the fader's tooltip names it, "Repeat Speed 1/8". The first note of a phrase still restarts the clock, so the clock's phase follows the playing, not the host's bar. It starts off. The web build's tempo is 120 BPM.

The matrix reads the clock through the **Gate** source, which is the gate the envelope sees, and **S&H** draws a new random value at each period while REPEAT is on. Its **Repeat Rate** destination moves SPEED (Section 9).

7 · GLIDE



1. **TIME** — portamento's time
2. **PORTAMENTO** — glides between notes
3. **BENDER** — a swoop up into each note

PORTAMENTO makes each voice glide from its previous pitch to the next, at **TIME**. It starts off. A voice glides from the last note it played: in **SOLO** that is the last note; in **POLY** a new note takes whichever voice is free, so the glide comes from that voice's last note.

TIME (SHORT to LONG) sets the glide's time constant: none at the foot, 0.2 s at the centre, where it starts, and 2 s at the top. A glide covers 95% of the way in three time constants: 0.6 s at the centre. The fader has an audio taper, so most of its travel is short times. A stack of more than one voice (Section 8) glides in 0.15 of the time, so a wide unison does not drag; the fader's readout gives the one-voice time.

The glide is exponential in frequency, not in pitch: it covers the same fraction of the distance in hertz in the same time. A rising glide therefore moves most of its pitch distance early, and a falling one spreads it more evenly: over an octave, when a rising glide has covered seven semitones a falling one has covered five.

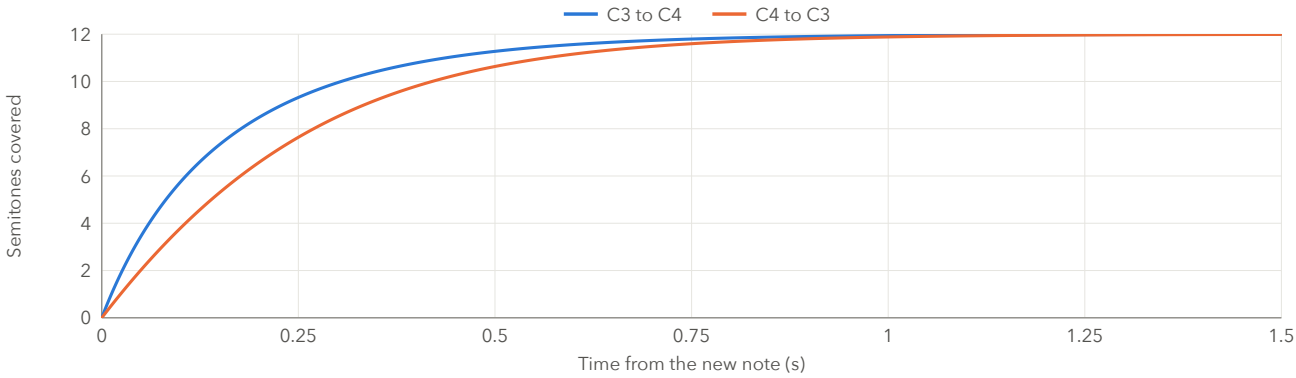


Figure 16. Portamento over an octave up and an octave down, TIME at its centre, SOLO, one voice.

BENDER starts each new note 20 Hz below its pitch and lets it rise into it, with a time constant of 80 ms: 95% of the way in 0.24 s. Like the vibrato, the depth is in hertz, so the swoop is about six semitones on C2, two on F♯3 and under one from C5 up. Turning BENDER off ends a swoop at once. With PORTAMENTO on as well, the swoop is glided too. It starts off.

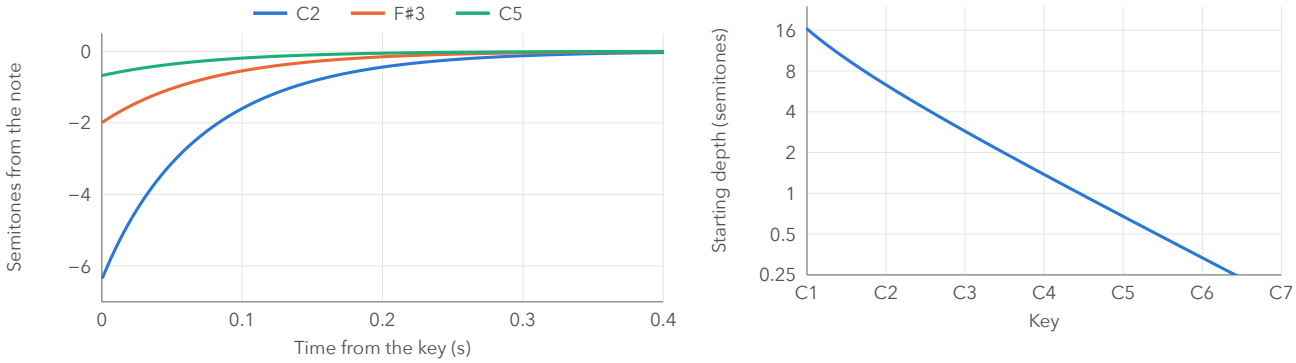
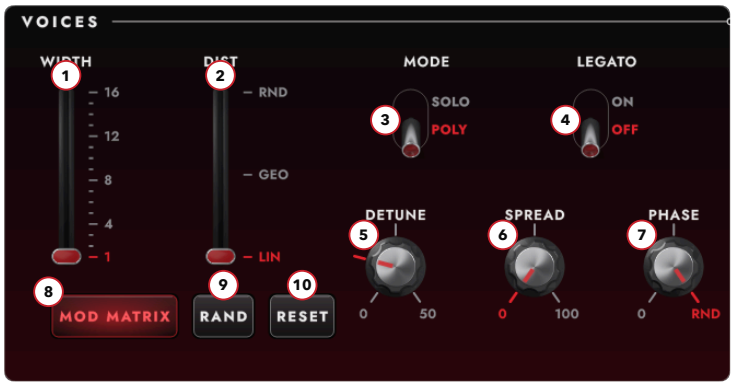


Figure 17. The Bender. Left: the swoop into three notes. Right: how far below the note each swoop starts, across the keyboard.

8 · VOICES



1	**WIDTH**	voices a note, 1 to 16	
2	**DIST**	how a stack's voices are spaced	
3	**MODE**	POLY or SOLO	
4	**LEGATO**	SOLO: no new strike when notes overlap	
5	**DETUNE**	a stack's detune, 0 to 50 cents	
6	**SPREAD**	the stack's Spread source, 0 to 100%	
7	**PHASE**	how far apart a stack's voices start	
8	**MOD MATRIX**	opens the matrix (Section 9)	
9	**RAND**	a new random sound (Section 12)	
10	**RESET**	back to Init	

Poly-Kritter has sixteen voices. **WIDTH** sets how many each note takes: at 1, each note is one voice and sixteen notes can sound; at 4, each note is a stack of four and four notes can sound; at 16, one note takes them all. Voices the division leaves over stay dark: no stack can claim them. The fader's tooltip says what the next note will do, "Width 2 stacks of 6 voices, 4 dark". A change of WIDTH takes effect at the next note.

WIDTH	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Notes	16	8	5	4	3	2	2	2	1	1	1	1	1	1	1	1
Dark	0	0	1	0	1	4	2	0	7	6	5	4	3	2	1	0

Poly and Solo

At **POLY** each note claims a stack. When none is free, a stack is taken from another note: released notes first, then held ones, oldest first. The note that loses its stack fades out over 5 ms rather than stopping dead, on four spare voices kept for the purpose, so a steal does not click. Playing a key again whose stack is still held re-strikes that stack.

At **SOLO** there is one stack, and the newest key plays; letting it go returns to the newest key still held. With **LEGATO** off, a new key re-strikes the envelope from where it is; with LEGATO on, a key played while another is held moves the pitch without a strike, so the envelope, the Bender and delay vibrato's fade carry on. SOLO at WIDTH 1 with LEGATO off is Kritter, the monophonic instrument.

The damper pedal (CC 64) holds notes let go under it, in both modes, until it comes up.

The stack

A stack is several copies of the whole voice on one note. **DETUNE** fans them out in pitch: the outermost sit at ±DETUNE and the rest between, so a wider stack is denser, not wider. **DIST** spaces them: **LIN** evenly, **GEO** gathered towards the outside, **RND** at random, drawn afresh for each note. DETUNE starts at 12 cents; it moves both VCOs alike, so a DUET stays in tune with itself.

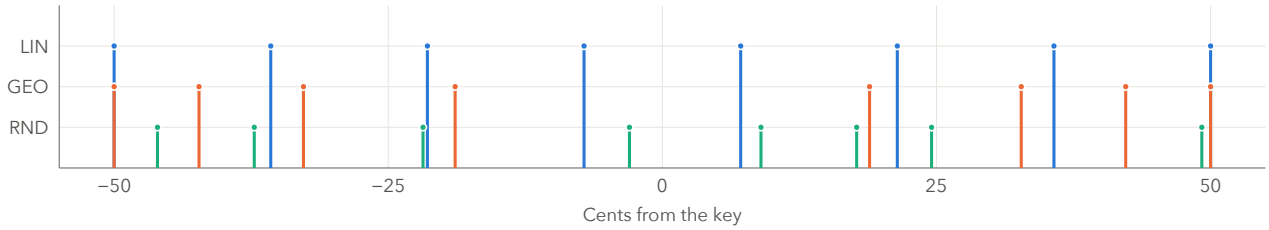


Figure 18. Where the eight voices of one note sound, WIDTH 8 and DETUNE 50, under each DIST: measured on each voice's pitch. RND's is one note's draw.

SPREAD gives each voice of a stack a value, its place in the stack from -1 to $+1$ times SPREAD, as the matrix's **Spread** source. It does nothing by itself. Init routes Spread to **Pan** in slot 1, so turning SPREAD up fans a stack across the stereo field; route it elsewhere and the outer voices of a stack sound different from the inner ones. It starts at 0.

PHASE sets how far apart the voices of a stack start their cycles: at 0 every voice starts in step and the attack is hard; at 100% (RND), where it starts, each starts at a random point. It applies to voices starting from silence. A single voice (WIDTH 1) is never reset, as Kritter's oscillator is not.

Level

The voices are summed at a quarter of their level, which leaves room for four notes: one note of SAW at C3 peaks at about -8 dBFS (Section 1), and a four-note chord about four times that. A stack of n voices is scaled by $1/\sqrt{n}$, so a wide stack is about as loud as a narrow one. Pan is constant-power, at unity in the centre: a voice panned hard to one side is 3 dB louder on that side.

9 · The modulation matrix

MOD MATRIX, on the VOICES plate, opens the matrix over the faceplate; its lamp is lit while any route is active, and its tooltip counts them. Click the cross, or outside the matrix, to close it.



Figure 19. The matrix, with Init's one route, Spread → Pan.

Each of the eight rows routes one **Source** to one **Dest** by an amount, **Depth**, from −100% to +100%, through a **Curve**, and optionally scaled by a second source, **Scale By**, through its own **Scale Curve**. Click a name to choose from its list, and drag the depth bar. **On** switches a row without losing it, and **×** clears it. Choosing a source on a blank row switches it on. A row is active when it is on with a depth other than zero; inactive rows are dimmed. The depths can be automated by the host; the rest of each row is part of the patch.

Every route is per voice: each voice reads its own sources and moves its own destinations. Routes to one destination add up. The faceplate's own routes (ENV, the vibrato switches, CUTOFF, the Bender, pitch bend) stay as they are and are not in the matrix.

Sources

Source	Range	Shared	What it is
Env	0 to 1		the voice's envelope, 0 to 1
LFO	±1	yes	the vibrato LFO's triangle
Velocity	0 to 1		the note's velocity, 0 to 1
Key	±1		the key, ±1 an octave from C4
Mod Wheel	0 to 1	yes	CC 1, 0 to 1
Aftertouch	0 to 1		the larger of poly and channel pressure
Note Random	0 to 1		0 to 1, drawn at each note
Spread	±1		the voice's place in its stack × SPREAD
Stack Pos	±1		the voice's place in its stack, −1 to +1
Gate	0 to 1		the gate the envelope sees, 0 or 1, Repeat's included
Voice Index	0 to 1		the voice's number, 0 to 1 across all twenty
Delay	0 to 1		delay vibrato's fade, 0 to 1 over 0.6 s
Pitch Bend	±1	yes	the bend, −1 to +1
Chorus LFO	±1		VCO 1's chorus sweep, a triangle
S&H	0 to 1		0 to 1, drawn at each note and each Repeat period
Constant	0 to 1	yes	1

A shared source has one value for every voice. **Velocity** and **Note Random** are taken at every note-on, a SOLO key over a held one included. Poly pressure resets at each note-on.

Destinations

Destination	At depth 100%	What it moves
Pitch	±12 semitones	both VCOs
VCO 2 Pitch	±12 semitones	VCO 2 alone, RING FIX's carrier included
Balance	±1	added to BALANCE
Low Pass	±4 octaves	the low-pass cutoff
High Pass	±4 octaves	the high-pass cutoff
Cutoff	±4 octaves	both cutoffs together
Reso	±1	added to RESO
Amp	±1	the VCA's level, in place of the envelope
Pan	±1	−1 left to +1 right
Attack	±1	added to ATTACK's position
Decay	±1	added to DECAY's position; at the top it holds
Env Length	±1	added to ATTACK and DECAY
Vibrato Depth	±1	added to vibrato DEPTH, before its taper
Filter Env	±1	added to the filter's ENV
VCO 1 Pitch	±12 semitones	VCO 1 alone
Vibrato Speed	±1	added to vibrato SPEED; shared
Repeat Rate	±1	added to Repeat SPEED; shared

Destinations that add to a control are in that control's own range: +1 takes a control from its foot to its top, and the sum is held within the control's range. The pitch destinations' depth is cubic, for fine control near zero: 50% is a semitone and a half.

Amp is the VCA's level. With nothing routed into it, it is the envelope, as on Kritter. Any active route into Amp replaces the envelope there, while the envelope still drives the filter's ENV: **Gate** into Amp gives an organ's on and off; **Env** into Amp scaled by **Velocity** gives velocity to level. In a route into Amp every source but Env and Gate reads 0 while the voice has no note, so an LFO into Amp is silent between notes.

Vibrato Speed and **Repeat Rate** move the one LFO and the one clock, so they take only shared sources (LFO, Mod Wheel, Pitch Bend, Constant). A route from a per-voice source into either is dimmed and does nothing. Under SYNC, Repeat Rate steps through the note values.

Curves and Scale By

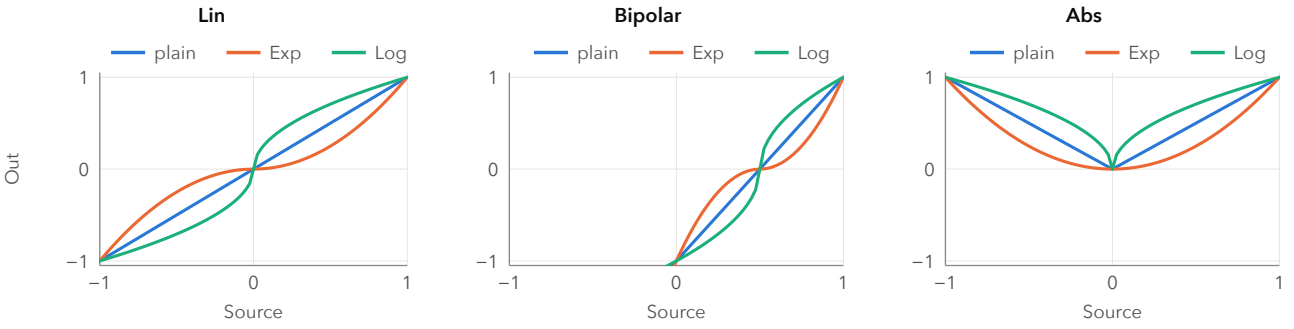


Figure 20. The nine curves. **Bipolar** turns a 0-to-1 source into −1 to +1, so velocity can push both ways about the middle; **Abs** folds a ±1 source to 0 to 1. **Exp** squares the value, gentle near zero and steep towards the ends; **Log** takes its square root, the reverse.

Scale By multiplies the route by a second source, read from 0 to 1 through its **Scale Curve**: a ±1 source is folded to 0 to 1 first. **LFO** into **Cutoff** scaled by **Mod Wheel** is vibrato on the filter that the wheel brings in; scaled by **Delay** it fades in with each note, as DELAY does for the pitch.

10 · FX

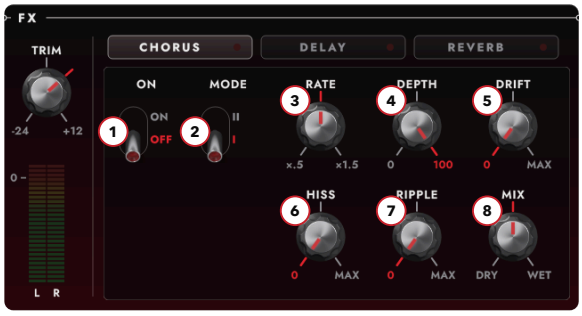


1	**TRIM**	level into the effects, −24 to +12 dB	2	**Meter**	the level after TRIM
3	**CHORUS**	the chorus page	4	**DELAY**	the delay page
5	**REVERB**	the reverb page			

The effects follow the voices in a fixed order, **TRIM**, chorus, delay, reverb, then **GAIN** and the limiter on the OUTPUT plate. Each effect has its own page with an **ON** switch, and the page's tab lights while it is on. Every effect starts off, and an effect that is off is out of the signal path entirely. Switching one on or off fades it over 30 ms.

TRIM sets the level into the effects, and the meter beside it shows that level; the chorus saturates gently when driven hard, so TRIM sets how much. It starts at 0 dB.

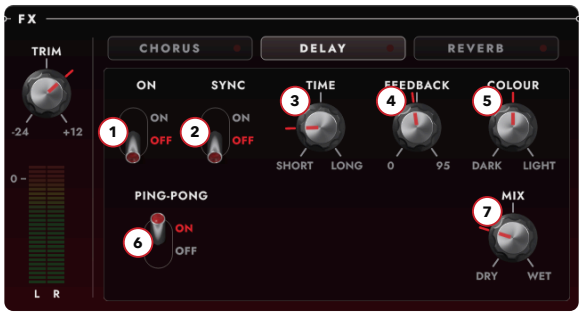
Chorus



1	**ON**	the chorus on
2	**MODE**	I or II
3	**RATE**	×0.5 to ×1.5 the mode's rate
4	**DEPTH**	the sweep's depth
5	**DRIFT**	a slow random wander
6	**HISS**	a bucket-brigade's noise
7	**RIPPLE**	a mains hum in the clock
8	**MIX**	dry to wet

A stereo bucket-brigade chorus. **MODE** I sweeps at 0.553 Hz and II at 0.898 Hz, at one depth; **RATE** scales either, ×1 at its centre. **DEPTH** sets how far the delay sweeps, from 1.66 to 5.35 ms at full, where it starts. **MIX** crossfades at equal power from dry to wet; it starts at 50%. The rest add an old unit's faults, each starting at 0: **HISS**, noise at the delay's input whose level follows the delay; **DRIFT**, a random wander of each side's delay by up to 10%; and **RIPPLE**, a 100 Hz wobble of both delays by up to 0.5%, as a mains hum on the clock gives.

Delay



1	**ON**	the delay on
2	**SYNC**	TIME as a note value
3	**TIME**	10 ms to 2 s, or a note value
4	**FEEDBACK**	repeats, 0 to 95%
5	**COLOUR**	darker or lighter repeats
6	**PING-PONG**	repeats alternate sides
7	**MIX**	dry to wet

A stereo delay. **TIME** runs from 10 ms to 2 s, 0.375 s at Init; with **SYNC** on it picks a note value of the host's tempo instead, from a whole note to a thirty-second triplet, 1/8. at Init. **FEEDBACK** (0 to 95%, 45% at Init) sets how many repeats; the loop is gently damped, so repeats soften as they go. **COLOUR** tilts each repeat about 1 kHz: towards DARK it cuts the highs, towards LIGHT the lows, by up to 12 dB a repeat; at the centre, where it starts, it does nothing. With **PING-PONG** on, as at Init, the repeats alternate between the sides. **MIX** starts at 25%.

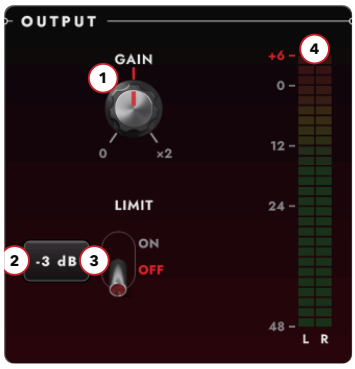
Reverb



1. **ON** — the reverb on
2. **SIZE** — the room's size
3. **DECAY** — 0.2 to 10 s
4. **DAMP** — how fast the highs die
5. **LOW CUT** — 20 Hz to 1 kHz, into the reverb
6. **MIX** — dry to wet

An eight-line feedback-delay reverb. **SIZE** sets the room's size (50% at Init) and **DECAY** its time, 0.2 s to 10 s, 2 s at its centre and 2.5 s at Init. **DAMP** (40%) makes the highs die sooner than the lows. **LOW CUT** keeps bass out of the reverb, 20 Hz to 1 kHz, 150 Hz at Init, so a bass note stays clear under a reverberant patch. **MIX** starts at 30%.

11 · OUTPUT and the scope



1. **GAIN** the output level, 0 to ×2
2. **−3 dB** sets GAIN from the last few seconds
3. **LIMIT** a safety limiter, last
4. **Meter** the output, −48 to +6 dBFS

GAIN scales the output, from silence at the left through unity at the centre, where it starts, to ×2 (+6 dB) at the right. Its tooltip shows it in decibels.

The **−3 dB** key sets GAIN so that the last few seconds' peak, measured before GAIN, sits at −3 dBFS. Play the sound as it will be played, then press. It is one press, one undo step: nothing keeps acting after it, and it does nothing below −60 dBFS. Random sounds (Section 12) are not level-matched, so this is the quick way to bring one into line.

LIMIT holds the output just under full scale, looking 2 ms ahead and letting go over 100 ms, so a big chord or a loud patch does not clip. It delays the output by 2 ms while on. It starts off.

The **scope** across the header shows the output's waveform, about 21 ms of it, starting at a rising zero crossing so a held note stands still. It has no controls and does not affect the sound.

MIDI and the host

Poly-Kritter plays notes and velocity, pitch bend (±2 semitones), the mod wheel (CC 1), the damper pedal (CC 64), and polyphonic and channel aftertouch. It follows the host's tempo for SYNC and the delay. Its output is stereo. The oscillators, effect section, filters and VCA run at twice the host's sample rate.

12 · Presets

The preset bar in the header shows the current preset's name, with a * once it has been changed; ‹ and › step through the presets, and a click on the name opens the browser of factory and user presets, by category. **SAVE** overwrites the current preset if it is a user preset, and otherwise acts as **SAVE AS**, which asks for a name and a folder. A preset holds every setting, the matrix and the effects included.

The browser's **Export** saves the current preset as a `.preset` file, or every user preset as one `.presetbank` file; **Import** reads either, as does dropping the file on the faceplate. So a sound found in the web build moves into the plugin: export it in the browser, then import the file in the plugin's. In the web build **Share** copies a link that opens the current sound. Import reads Poly-Kritter's presets made by the same version, or one that differs only in its last number; it refuses another instrument's.

The factory bank holds eighty sounds, ten in each of eight categories: Bass, Brass, FX, Keys, Lead, Pad, Pluck and Poly. Each began as a random sound of its kind (below), kept by ear; the name's tooltip gives its kind and seed.

RAND, on the VOICES plate, makes a new sound, as one undo step. It picks one of the eight kinds and builds the whole patch in that kind's manner:

Kind	Mode	Character
Bass	Solo	16' or 32', square or saw, Key → Cutoff, dry or a short delay
Keys	Poly	8', plucky to sustained, velocity to the filter
Poly	Poly	8' or 16', brighter and sustained, chorus likely
Brass	Poly	saw, SUSTAIN on, a strong filter envelope, sometimes delayed vibrato
Pad	Poly	slow attack, LONG and SUSTAIN on, slow movement, reverb likely
Lead	Solo or Poly	8' or 4', aftertouch or wheel vibrato, delay likely, glide sometimes
Pluck	Solo	SUSTAIN off, a short DECAY, a high ENV
FX	Solo	anything: ring modes, noise, REPEAT, SYNC, a wild matrix

Every kind but FX gets at least one route for playing: the mod wheel, velocity or aftertouch. Stacks are narrow in POLY, so that chords have voices to spare, and may be wide in SOLO. **GAIN**, **TRIM** and **LIMIT** stay as they were. The sound gets a two-word name, and the name's tooltip its kind and a five-character seed, "Random Pad, seed 3F2AK": quote the seed to report a sound. **SAVE AS** then offers a user folder named after the kind. Pressing RAND twice rarely gives the same kind twice running: when it does, it redraws half the time.

RESET returns every setting to Init, as one undo step.

13 · Patches

Six factory presets, each chosen to show one part of the instrument. Load each by name. Each is drawn as it loads, with its matrix routes.

Spindly Sieve

Keys · Random Keys, seed 65SK6

RING FIX. A struck, bell-like key. VCO 2 sits at ×4 under RING FIX, so the ring modulator's carrier stays at 440 Hz whatever key is played, and every note gets its own inharmonic set of sidebands: the note plus and minus 440 Hz, and so on. **BALANCE** at 42% keeps some of the plain sawtooth underneath, so the notes still have a pitch. Two voices a note, detuned and spread, then the chorus at MODE II and a short reverb.



Figure 21. Spindly Sieve, as it loads.

Slot	Source	Dest	Depth	Curve	Scale By	Scale Curve
1	Spread	Pan	100%	Lin	None	
2	Note Random	Pitch	15%	Bipolar	None	
3	Velocity	Cutoff	41%	Lin	None	

Different from Init: Select Ring Fix, Balance 42 %, VCO 2 24.00 st, Tuning Equal, Low Pass 658 Hz, Reso 6 %, Env 38 %, Attack 0.003 s, Decay 1.606 s, Sustain On, Gain -5.7 dB, Width 2, Detune 19.1 ct, Spread 89 %, Chorus On On, Chorus Mode II, Chorus Rate 0.92 ×, Chorus Depth 90 %, Chorus Mix 36 %, Chorus Hiss 1 %, Chorus Drift 20 %, Chorus Ripple 7 %, Reverb On On, Reverb Size 51 %, Reverb Decay 2.456 s, Reverb Damp 54 %, Reverb Mix 24 %.

Austere Bookcase

Pluck · Random Pluck, seed FQ35Y

The filter envelope and velocity. A pluck: the low-pass sits at 175 Hz and **ENV** sweeps it up at each strike, while **DECAY** below HOLD lets each note die even with the key held, and SUSTAIN off stops it with the key. **Velocity** opens the cutoff and deepens **Filter Env**, so a hard note is brighter twice over. **SOLO**, eight voices a note.



Figure 22. Austere Bookcase, as it loads.

Slot	Source	Dest	Depth	Curve	Scale By	Scale Curve
1	Spread	Pan	100%	Lin	None	
2	Velocity	Filter Env	39%	Lin	None	
3	Velocity	Cutoff	47%	Lin	None	

Different from Init: Wave Square, Balance 50 %, VCO 2 7.02 st, Tuning Equal, Low Pass 175 Hz, Reso 15 %, Env 79 %, Attack 0.002 s, Decay 0.362 s, Gain 4.4 dB, Width 8, Mode Solo, Detune 12.1 ct, Spread 47 %, Dist Geo.

Quaint Bowl

Lead · Random Lead, seed V7H2D

SOLO and LEGATO. A lead at 4', one voice. With **LEGATO** on, a key played while another is held slides to the new pitch without a strike: play a line without letting go and the envelope and the delayed vibrato carry on through it; lift between notes and each note strikes, and its vibrato fades in afresh over 0.6 s. **Note Random** nudges each note's pitch by a few cents, and the mod wheel opens the filter.



Figure 23. Quaint Bowl, as it loads.

Slot	Source	Dest	Depth	Curve	Scale By	Scale Curve
1	Spread	Pan	100%	Lin	None	
2	Note Random	Pitch	15%	Bipolar	None	
3	Velocity	Cutoff	33%	Lin	None	
4	Mod Wheel	Cutoff	35%	Lin	None	

Different from Init: Scale 4', Balance 32 %, VCO 2 3.16 st, Tuning Just, Low Pass 2586 Hz, Reso 29 %, Env 42 %, Attack 0.006 s, Decay 136.071 s, Sustain On, Vibrato Speed 4.58 Hz, Vibrato Depth 0.54 Hz, Delay Vibrato On, Gain -2.5 dB, Mode Solo, Legato On.

Kindly Sandwich

FX · Random FX, seed 6RG20

REPEAT and SYNC. One held key plays a stream of eighth notes at the host's tempo: **REPEAT** re-strikes the envelope from the synced clock, and SUSTAIN on lets each strike ring into the next. The voice is RING FIX against a 4' CH II pulse with **BALANCE** nearly full, so each strike clangs; the Bender scoops into each new key, and the delay echoes the stream.



Figure 24. Kindly Sandwich, as it loads.

Slot	Source	Dest	Depth	Curve	Scale By	Scale Curve
1	Spread	Pan	100%	Lin	None	
2	Constant	Filter Env	-98%	Lin	Constant	Exp
3	Env	Low Pass	-20%	Abs Log	None	
4	Chorus LFO	VCO 2 Pitch	-37%	Abs Log	Chorus LFO	Lin

Different from Init: Scale 4', Wave Chorus II, Select Ring Fix, Balance 94 %, VCO 2 3.86 st, Fine -40.0 ct, Tuning Equal, High Pass 21 Hz, Low Pass 1253 Hz, Reso 31 %, Env 83 %, Attack 0.110 s, Decay held, Sustain On, Vibrato Speed 6.92 Hz, Vibrato Depth 9.08 Hz, Vibrato On, Repeat Speed 9.31 Hz, Repeat On, Sync On, Bender On, Gain -8.8 dB, Width 8, Mode Solo, Detune 14.3 ct, Spread 55 %, Dist Rnd, Delay On On, Delay Time 0.231 s, Delay Feedback 49 %, Delay Mix 25 %, Delay Colour 0.24.

Eerie Muffin

Bass · Random Bass, seed WBZ9G

The stack. A bass of sixteen voices on one note: **SOLO**, **WIDTH** 16, detuned by up to 24 cents at random spacing (**RND**) and spread wide by **SPREAD** through slot 1's Spread → Pan. One key fills the stereo field. **Key** into **Cutoff** brightens the high notes, and the mod wheel opens the filter further. Set WIDTH to 1 to hear a single voice of it.



Figure 25. Eerie Muffin, as it loads.

Slot	Source	Dest	Depth	Curve	Scale By	Scale Curve
1	Spread	Pan	100%	Lin	None	
2	Key	Cutoff	24%	Lin	None	
3	LFO	Cutoff	6%	Lin	None	
4	Mod Wheel	Cutoff	38%	Lin	None	

Different from Init: Scale 16', Wave Triangle, Balance 22 %, VCO 2 20.84 st, Tuning Just, Low Pass 550 Hz, Reso 44 %, Env 54 %, Attack 0.002 s, Decay 2.281 s, Sustain On, Gain -0.4 dB, Width 16, Mode Solo, Detune 23.8 ct, Spread 86 %, Dist Rnd.

Rotund Ladder

Pad · Random Pad, seed 4MNDX

LONG and the mod wheel. A pad. **LONG** stretches the envelope eleven times: the attack takes about a second, and SUSTAIN's release lingers for seconds more. VIBRATO is on with **DEPTH** at 0, so there is no vibrato until the mod wheel adds it through **Vibrato Depth**. The chorus LFO moves the cutoff, the vibrato LFO pans each voice, and chorus and reverb finish it.



Figure 26. Rotund Ladder, as it loads.

Slot	Source	Dest	Depth	Curve	Scale By	Scale Curve
1	Spread	Pan	100%	Lin	None	
2	Chorus LFO	Cutoff	14%	Lin	None	
3	LFO	Pan	44%	Lin	None	
4	Mod Wheel	Vibrato Depth	36%	Lin	None	

Different from Init: Wave Triangle, Balance 40 %, VCO 2 7.02 st, Fine 1.4 ct, Tuning Equal, High Pass 109 Hz, Low Pass 2614 Hz, Reso 22 %, Env 18 %, Attack 0.088 s, Decay 7.276 s, Sustain On, Long On, Vibrato Speed 2.16 Hz, Vibrato Depth 0.00 Hz, Vibrato On, Gain -14.6 dB, Width 3, Detune 14.2 ct, Spread 76 %, Dist Rnd, Chorus On On, Chorus Rate 1.18 ×, Chorus Depth 77 %, Chorus Mix 47 %, Chorus Hiss 3 %, Chorus Drift 10 %, Chorus Ripple 5 %, Reverb On On, Reverb Size 78 %, Reverb Decay 5.293 s, Reverb Damp 23 %, Reverb Mix 38 %.

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Hertz, not pitch

Kritter's vibrato, Bender and portamento act on the pitch in hertz, not in semitones, as the instrument they model did. A fixed deviation in hertz is a large interval on a low note and a small one on a high note, and an exponential glide in hertz is not symmetrical in pitch:

- Vibrato at one DEPTH is deeper on low notes: it halves with each octave up.
- The Bender's swoop is wide on low notes and slight on high ones.
- Portamento covers more of its distance early when rising than when falling.

The hertz are counted at 8', so SCALE moves a note and its vibrato together: a key at 4' has the vibrato, in cents, of the same key at 8'. For vibrato that is even across the keyboard, route **LFO** to **Pitch** in the matrix; the matrix's pitch routes are in semitones.

Tapers

Control	Foot	¼	½	¾	Top
ATTACK, to the peak	0.002 s	0.009 s	0.045 s	0.211 s	1 s
DECAY, to −40 dB	0.7 s	1.6 s	5.5 s	44.2 s	held
Vibrato SPEED	2 Hz	3.31 Hz	5.48 Hz	9.06 Hz	15 Hz
Vibrato DEPTH	0 Hz	0.68 Hz	2.7 Hz	8.78 Hz	27 Hz
Repeat SPEED	2 Hz	3.76 Hz	7.07 Hz	13.3 Hz	25 Hz
GLIDE TIME, to 95%	0 s	0.15 s	0.6 s	1.95 s	6 s
HIGH/LOW PASS	20 Hz	113 Hz	640 Hz	3620 Hz	20480 Hz
Reverb DECAY	0.2 s	0.63 s	2 s	4.47 s	10 s
Reverb LOW CUT	20 Hz	53 Hz	141 Hz	376 Hz	1000 Hz

LONG multiplies ATTACK and DECAY by eleven. The filter figures are at 8'. Vibrato DEPTH and GLIDE TIME share the audio taper: a tenth of full value at half travel.

Note values

Repeat SPEED, SYNC	Beats	Hz at 120 BPM
4/1	16	0.125
2/1	8	0.250
1/1.	6	0.333
1/1	4	0.500
1/2.	3	0.667
1/1T	2.667	0.750
1/2	2	1.000
1/4.	1.5	1.333
1/2T	1.333	1.500
1/4	1	2.000
1/8.	0.75	2.667
1/4T	0.667	3.000
1/8	0.5	4.000
1/16.	0.375	5.333

Delay TIME, SYNC	Beats	ms at 120 BPM
1/1	4	2000.0
1/1.	6	3000.0
1/1T	2.667	1333.3
1/2	2	1000.0
1/2.	3	1500.0
1/2T	1.333	666.7
1/4	1	500.0
1/4.	1.5	750.0
1/4T	0.667	333.3
1/8	0.5	250.0
1/8.	0.75	375.0
1/8T	0.333	166.7
1/16	0.25	125.0
1/16.	0.375	187.5

1/8T	0.333	6.000
1/16	0.25	8.000
1/32.	0.188	10.667
1/16T	0.167	12.000
1/32	0.125	16.000
1/32T	0.083	24.000

1/16T	0.167	83.3
1/32	0.125	62.5
1/32.	0.188	93.8
1/32T	0.083	41.7

Repeat's twenty note values each take an equal share of SPEED's travel, slowest at the foot. A dot is half as long again, T a triplet.

Ranges

Control	Range	Init
SCALE	32' to 2'	8'
VCO 2	×1 to ×4	×1
FINE	±50 cents	0
RING FIX carrier	110 to 440 Hz	—
HIGH PASS, LOW PASS	20 Hz to 20.5 kHz, an octave a mark	0, 10
Filter Q	0.71 (OFF); rings ≈5 cycles (LP), ≈4 (HP) at ON	OFF
ENV	up to +4 octaves at the envelope's peak	OFF
ATTACK	2 ms to 1 s; ×11 with LONG	FAST
DECAY	τ 0.15 s to held; ×11 with LONG	HOLD
Release	τ 10 ms; 0.3 s with SUSTAIN, ×11 with LONG	10 ms
Vibrato SPEED	2 to 15 Hz	5.5 Hz
Vibrato DEPTH	0 to ±27 Hz	±2.7 Hz
CUTOFF vibrato	±0.4 octave at DEPTH's centre, ±4 at full	off
Delay vibrato	0.6 s fade-in	off
Repeat SPEED	2 to 25 Hz; 4/1 to 1/32T with SYNC	7.1 Hz
GLIDE TIME	τ 0 to 2 s; ×0.15 for a stack	0.2 s
Bender	−20 Hz, τ 80 ms	off
WIDTH	1 to 16 voices a note	1
DETUNE	0 to 50 cents	12
Voices	16 sounding, 4 more for steals	
Matrix	8 routes	Spread → Pan
TRIM	−24 to +12 dB	0 dB
GAIN	0 to ×2 (+6 dB)	×1
Pitch bend	±2 semitones	
Oversampling	2×, oscillators to VCA	

Troubleshooting

No sound from the effect section BALANCE is at 0, or EFFECT is off.

The sound goes quiet as a filter fader moves STOP is off and LOW PASS is below HIGH PASS, or a matrix route to High Pass has lifted it past the notes.

A held note dies away DECAY is below HOLD. A note that stops with the key: SUSTAIN is off.

Vibrato is too wide on low notes and too narrow on high ones see Hertz, not pitch, above.

VIBRATO is on but there is no vibrato DEPTH is at 0, waiting for the matrix (the mod wheel or aftertouch into Vibrato Depth).

Only a few notes sound at once WIDTH is high. Sixteen voices divide by WIDTH; the tooltip on WIDTH says how many notes can sound.

A glide is barely heard the patch stacks voices, and a stack glides in 0.15 of TIME. Raise TIME, or set WIDTH to 1.

A route does nothing it may be off or at depth 0, or dimmed because a per-voice source cannot drive Vibrato Speed or Repeat Rate.

Velocity does nothing Kritter has no velocity; route it in the matrix.

A random sound is too loud or too quiet play it, then press −3 dB; or switch LIMIT on.